

NETHRAJA KANDULA

Email: nethrajakandula@gmail.com

Phone: +1 (681) 350-2371

LinkedIn: www.linkedin.com/in/nethraja-kandula

Food Science researcher with expertise in food microbiology, molecular diagnostics, digital PCR, whole-genome sequencing, and pathogen detection in poultry and seafood systems. Experienced in assay development, microbial growth analysis, bioinformatics, teaching, mentoring, and translating research into applied food safety solutions.

EDUCATION

Florida State University, Tallahassee, FL

August 2022 – July 2026

Doctorate (Ph.D.): Nutrition and Food Science

Current CGPA: 3.58/4

Osmania University, Telangana, India

August 2017 – July 2021

Bachelor of Technology (B. Tech.): Food Technology

Overall CGPA: 8.2/10

RESEARCH AND TEACHING EXPERIENCE

Research Assistant

January 2023 – present

Florida State University, Tallahassee, Florida

- Develop and standardize dPCR assays for the detection and quantification of *Salmonella* in poultry products (whole carcass rinses, chicken breast, and ground turkey).
- Characterizing the growth curves of various *Salmonella* strains.
- Whole-genome sequencing of various *Salmonella* strains to create a CRISPR database.
- Hybrid-capture-based Next-Generation Sequencing (NGS) approach for serotyping of *Salmonella* in poultry products.

Teaching Assistant

January 2024 – December 2025

Florida State University, Tallahassee, Florida

- Teach scientific principles of cooking techniques to help students understand how ingredients and processes contribute to the final dish.
- Guide students through preparing diverse recipes, providing hands-on assistance, and feedback to ensure culinary accuracy and safety.
- Designed and implemented course syllabus with innovative assignments that help students reflect on the understanding of the science in the lab.

Research Intern

March 2022 – June 2022

Samath Global Food Consultants, Telangana, India

- Formulated seasoning mixes for potato chips and developed protein powder flavor variants.

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TRAINING AND CERTIFICATIONS

- IHA-accredited course on “**Certified HACCP Principles and GMP Training for Food Processors and Manufacturers**” (January 2025) by the Department of Food Science and Technology, University of Georgia.
- **National Science Foundation (NSF) Innovation Corps (I-Corps™)** Mid-South Hub Regional Sullivan Family Ideator I-Corps Cohort (August-October 2024) by The Wond’ry, Nashville, Tennessee, USA.
- **Teaching Assistant Course Design** Institute (July 2024) by the Center for the Advancement of Teaching, Florida State University, USA.
- State-level E-conference on “**Emerging Trends in Food Processing**” (August 2021) by Dept. of Food Processing and Engineering, DDU Kashal Kendra, JSS College of Arts, Commerce and Science, Karnataka, India.
- Online training program on “**HACCP and Food Allergens Management –Level 1**” (August 2021) by EP Infoways Institute of Food Safety & Hygiene, Noida, India.
- **Student Intern** at Little Star Foods Pvt. Ltd., (May 2019) Hyderabad, Telangana, India.

GRANTS

- Singh, P., Kwawukume, S., **Kandula, N. (Co-I)**. Quantifying Florida’s Seafood Fraud: High-Resolution dPCR Assessment of Imported Shrimp Blending in the Domestic Supply Chain. Southern Shrimp Alliance. (Submitted)
- Singh, P., Bosilevac, J. M., Walker, L., **Kandula, N. (Co-I)** (April 2025– April 2026). Digital PCR method for the detection and estimation of *Salmonella* load in poultry samples. USPoultry. Total award: \$79,763
- **Kandula, N. (PI)**, Singh, P. (October 2024– July 2025) NSF I-Corps Hub: Mid-South Region Microgrant. Total award: \$2,500

AWARDS

- Finalist, **Three Minute Thesis Competition**, Florida State University, 2026.
- **IFT Innovation Award** for Spring 2026.
- **Jean A. Reutlinger & Lillian H. Munn Scholarship** for Fall 2024/Spring 2025.
- **Dr. Shridhar Sathe Graduate Scholarship and Research Fund** for Fall 2024/Spring 2025.
- **Walter F. Hayes Family Endowment Scholarship** for Fall 2023 and Spring 2024.

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PROFESSIONAL SERVICE AND LEADERSHIP

Volunteer, Three Minute Thesis Competition, International Association for Food Protection (IAFP) Annual Meeting (2026)

- Supported organization and on-site coordination of the Three Minute Thesis Competition during the IAFP Annual Meeting.
- Assisted with participant support, event logistics, and communication to help ensure smooth execution of the competition.

Volunteer, IFT Florida Suppliers Night (2025, 2026)

- Supported event operations, including attendee guidance, exhibitor assistance, and on-site logistics to ensure smooth execution of the professional networking program.
- Assisted with real-time coordination and troubleshooting during the event to maintain an organized experience for participants.

Student Volunteer, IFT Annual Expo, Food Microbiology Division (2024–2025)

- Supported Food Microbiology Division activities during the IFT Annual Expo, assisting with session/event logistics and participant engagement.
- Contributed to division outreach by supporting communication and attendee coordination during scheduled activities.

Mentored/Trained Undergraduate Researchers in Core Lab Methods (2025)

- Trained undergraduate researchers in foundational lab techniques and good laboratory practices, emphasizing aseptic technique, accurate documentation, and laboratory safety.
- Provided hands-on guidance in experimental workflow execution and troubleshooting to promote reproducibility and research independence.

Coordinated and Delivered On-Site Assay Demonstrations at Federal Laboratories, Alabama (2024)

- Coordinated and conducted on-site demonstrations of a molecular assay developed for the identification of imported shrimp species.
- Prepared demonstration materials and explained the end-to-end workflow (sample handling considerations, assay setup, and result interpretation) to support adoption and consistent use by laboratory personnel.

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CONFERENCE ABSTRACTS

- **Kandula, N.**, Kwawukume, S., Singh, P. (2025). Shelf-life and Microbial Community Dynamics of Reduced Oxygen Packaged Seafood Products. International Association for Food Protection (IAFP) 2025 Annual Meeting, Huntington Convention Center of Cleveland, Ohio, USA, July 27 – July 30.
- **Kandula, N.**, Brown, H., Kwawukume, S., Singh, P. (2024). Rapid RNase H-dependent PCR Lateral Flow Assay for the Detection of Pacific Whiteleg Shrimp, *Litopenaeus vannamei*. Institute of Food Technologists (IFT) 2024 Annual Event and Expo, McCormick Place, Illinois, USA, July 14 – July 17.
- **Kandula, N.**, Velez, F. J., Bosilevac, J. M., & Singh, P. (2023). Digital PCR Assay for the Specific Detection and Estimation of *Salmonella* Contamination Levels in Poultry Rinse. Florida Association of Food Protection (FAFP) 2023 Annual Meeting, Lake Buena Vista, Florida, USA, May 16 – May 18.

PUBLICATIONS

- **Kandula, N.**, Kwawukume, S., Williams, D., Singh, P. (2025) Rapid RNase H-dependent PCR (rhPCR) Lateral Flow Assay for the Detection of Pacific White Shrimp, *Litopenaeus vannamei*. Journal of Food Composition and Analysis. Volume 144, 2025, 107648, ISSN 0889-1575, <https://doi.org/10.1016/j.jfca.2025.107648>.
- Cheng, J., **Kandula, N.**, Cortes, V. E., Choi, H., Singh, P., & Cui, L. (2024). Pea protein isolate-based active films for salmon preservation: The role of different essential oils in film properties, antioxidant, and antibacterial activities. Current Research in Food Science, 9, 100936. <https://doi.org/10.1016/j.crfs.2024.100936>
- **Kandula, N.**, Velez, F. J., Jackson, C. R., Bosilevac, J. M., & Singh, P. (2024). Varying amplification efficiency real-time PCR assay for the specific detection and estimation of *Salmonella* contamination levels in poultry rinse. Food Bioscience, 61, 104777. <https://doi.org/10.1016/j.fbio.2024.104777>
- Velez, F. J.*, **Kandula, N.***, Blech-Hermoni, Y., Jackson, C. R., Bosilevac, J. M., & Singh, P. (2024). Digital PCR assay for the specific detection and estimation of *Salmonella* contamination levels in poultry rinse. Current Research in Food Science, 9, 100807. <https://doi.org/10.1016/j.crfs.2024.100807> (*Equal contribution)
- Kwawukume, S., Velez, F. J., **Kandula, N.**, Cui, L., & Singh, P. (2023). Rapid RNase H-dependent PCR lateral flow assay for the detection of red snapper. Food Bioscience, 103390. <https://doi.org/10.1016/j.fbio.2023.103390>

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Under Review

- Kwawukume, S., **Kandula, N.**, Cui, L., Singh, P. (2026). A Rapid rhPCR–Lateral Flow Assay for Authenticating Royal Red (*Pleoticus robustus*) and Argentine Red Shrimp (*Pleoticus muelleri*) in the U.S. Seafood Supply Chain. *Journal of Food Science and Biotechnology*.
- **Kandula, N.**, Cortes, V. E., Brown, H., Blech-Hermoni, Y., Jackson, C. R., Bosilevac, J. M., & Singh, P. (2026). Rapid Quantification of *Salmonella* in Chicken Breast Using Extraction-Free Digital PCR Following Short Enrichment. *Journal of Poultry Science*.
- Cortes, V. E., Brown, H., **Kandula, N.**, Mishra, A., Jackson, C. R., Bosilevac, J. M., & Singh, P. (2026). Extraction-Free Digital PCR for Rapid Absolute Quantification of *Salmonella* in Ground Beef. *Journal of Food Control*.
- **Kandula, N.**, Kwawukume, S., Cortes, V. E., Cheng, J., Cui, L., Singh, P. (2026). Shelf-life and Microbial Community Dynamics of Ten Vacuum Skin Packed (VSP) Seafood Products. *International Journal of Food Microbiology*. *LWT Journal*.

MEDIA COVERAGE AND UNIVERSITY PROFILES

- **“Nethraja Kandula.”** *Florida State University Office of Graduate Fellowships and Awards – Grad Impact Profile*, 2026. Featured graduate student profile highlighting doctoral research in food safety, digital PCR-based *Salmonella* quantification, NSF I-Corps participation, research translation, and FSU Three Minute Thesis finalist recognition. <https://ogfa.fsu.edu/graduate/nethraja-kandula>
- **“Something’s fishy: Florida State students test local shrimp.”** *Tallahassee Democrat*, March 7, 2026. Featured coverage of our shrimp mislabeling/authenticity research at Florida State University. <https://www.tallahassee.com/story/news/local/state/2026/03/07/bait-and-switch-fsu-researchers-dive-into-local-seafood-mislabeling/88862550007/>
- **“Research to Impact: Highlighting the journey of one FSU graduate student through the National Science Foundation’s I-Corps program.”** *Florida State University News*, May 7, 2025. Featured story on research translation and participation in the NSF I-Corps program. <https://news.fsu.edu/news/science-technology/2025/05/07/research-to-impact-highlighting-the-journey-of-one-fsu-graduate-student-through-the-national-science-foundations-i-corps-program/>
- **Shrimp testing at FSU (photo gallery).** *Tallahassee Democrat*, February 14, 2025. Photo coverage highlighting shrimp testing research activities at Florida State University. <https://www.tallahassee.com/picture-gallery/news/2025/02/14/shrimp-testing-fsu/78454830007/>

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SKILLS

Technical Skills: Microbial culture, selective plating, colony enumeration, DNA extraction, qPCR, dPCR, rhPCR, lateral flow assay development, whole-genome sequencing, CRISPR analysis, microbial growth modeling.

Bioinformatics/Data Analysis: CLC Genomics Workbench, DRAGEN, SeqSero2, CRISPRCasFinder, Python, MATLAB, GraphPad Prism, SPSS.

Professional Skills: Scientific writing, manuscript preparation, teaching, mentoring, grant writing, science communication.

KEY STRENGTHS

- Led and motivated a team of laboratory research assistants to troubleshoot and resolve assay inconsistencies, improving workflow efficiency.
- Effectively managed multiple projects concurrently, balancing laboratory work, data analysis, and scientific reporting under tight deadlines.
- Adapted protocols swiftly to address unexpected methodological challenges, minimizing disruptions in experimental progress.
- Excellent communication abilities, including scientific writing, editing research manuscripts, and delivering presentations.
- Analytical thinker with proficiency in quantitative data analysis, modeling microbial growth kinetics, and experimental design.
- Detail-oriented in data documentation, ensuring accuracy, reproducibility, and compliance with regulatory standards.